

Why did Polls Miss? (in 2020)

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Polling error is often just treated as noise. But is it random, or does it follow predictable patterns?

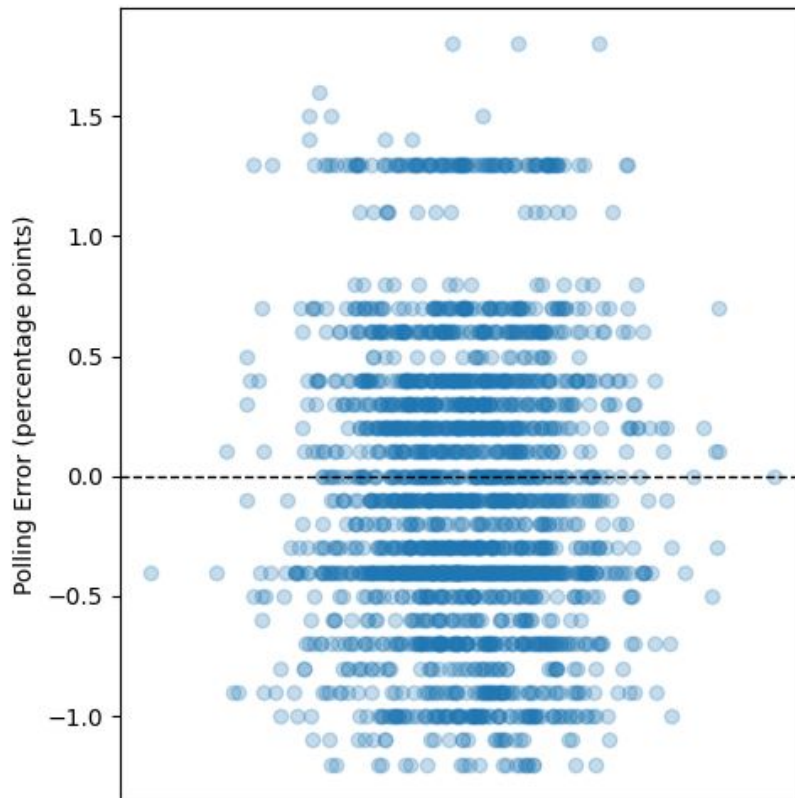
What questions does this analysis answer?

- Is polling error inherently random, or structured?
- How polling error evolves as Election Day approaches
- The role of partisanship in polling error
- What pollster quality actually predicts
- Is underestimation systemic or pollster-specific?
- Why quality does **not** prevent major polling misses

Distribution of signed polling error across 2020 polls



Polling Error Is Widely Dispersed



1. Errors are Large

Polling errors are often several percentage points in magnitude, with many polls missing by more than 1 point.

2. Errors are not tightly centered around zero

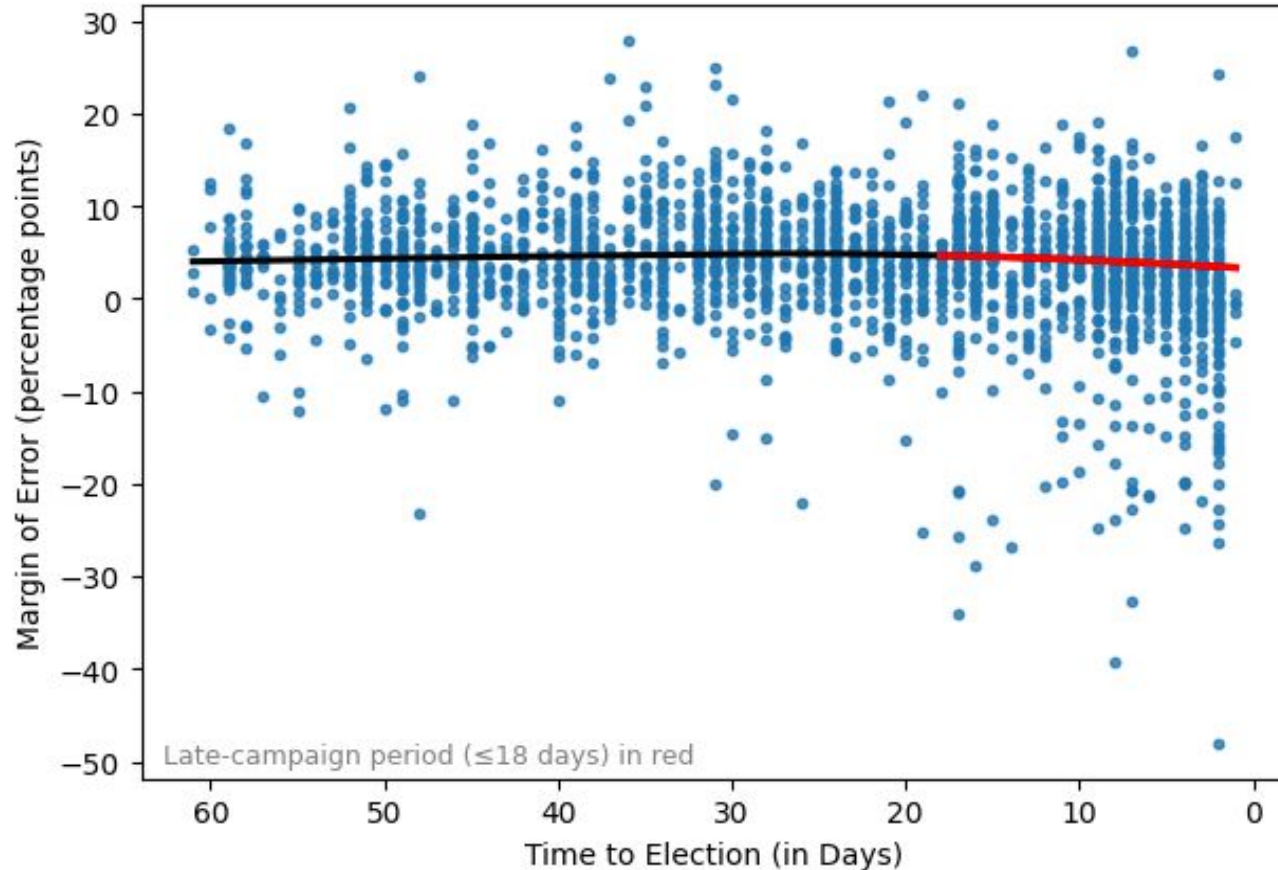
This might suggest that polling outcomes are not dominated by small random fluctuations.

3. Errors are asymmetric and heavy-tailed

The distribution shows substantial asymmetry and long tails, indicating that there may be structured variation rather than simple noise.

So, if polling error doesn't exhibit a typical random distribution, how does it change over the course of the election cycle?

Polling Error Evolves as Election Day Approaches



1. Polling error is not stable over time

The magnitude and direction of polling error vary systematically as Election Day approaches, rather than remaining constant throughout the campaign.

2. Late-campaign polls show increased asymmetry

Errors near Election Day are more unevenly distributed, suggesting that late polling may behave differently from earlier polling.

Points represent individual polls; line shows a LOESS-smoothed trend.

If polling error changes over time, the next question is whether it differs by who conducts the poll.



Immediately, a few options might jump into mind.

1

Partisan
Incentives

2

Pollster
Quality

3

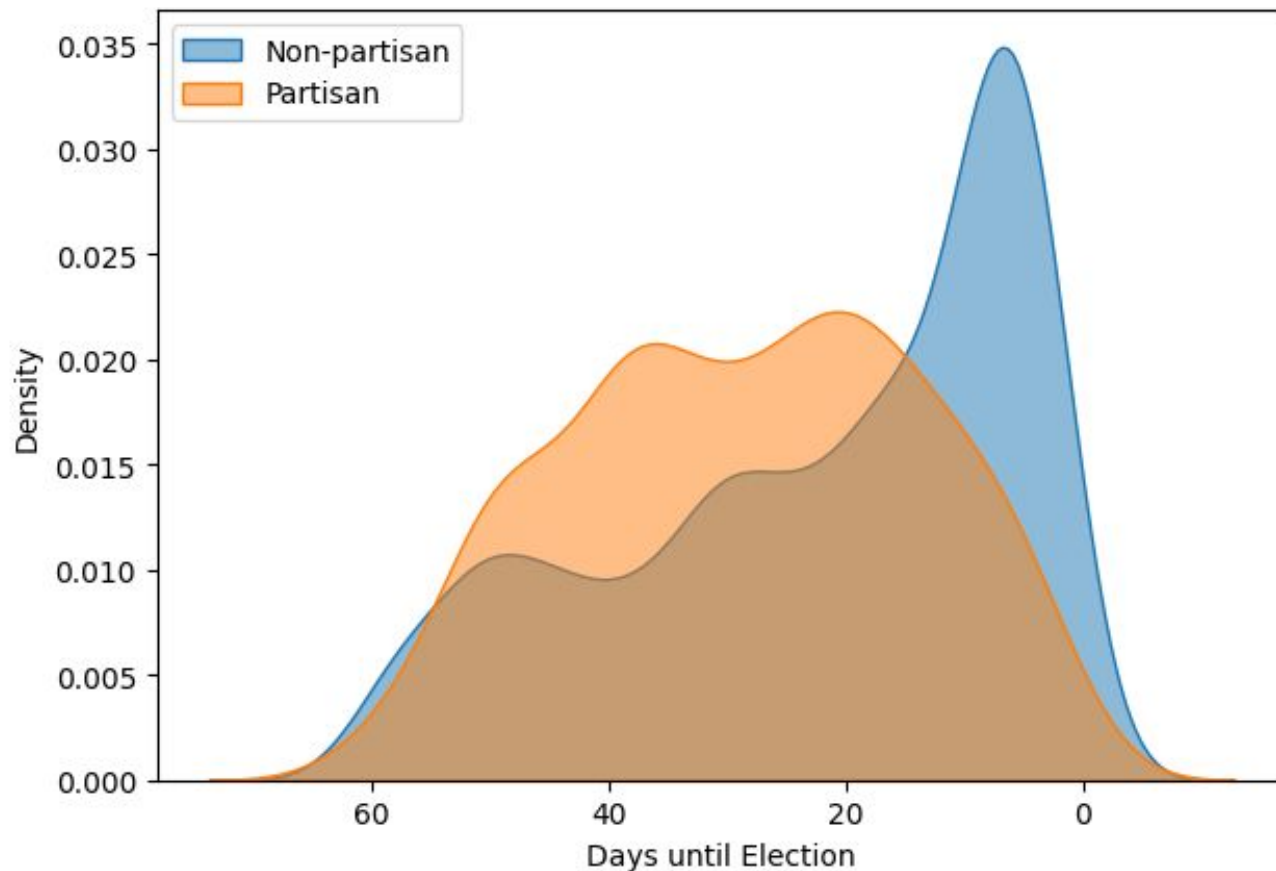
Election-wide
dynamics

4

Future
Directions

1. Partisan Incentives

Partisan, Nonpartisan Polls Are Conducted at Different Times



1. Partisan and nonpartisan polls follow different timing patterns

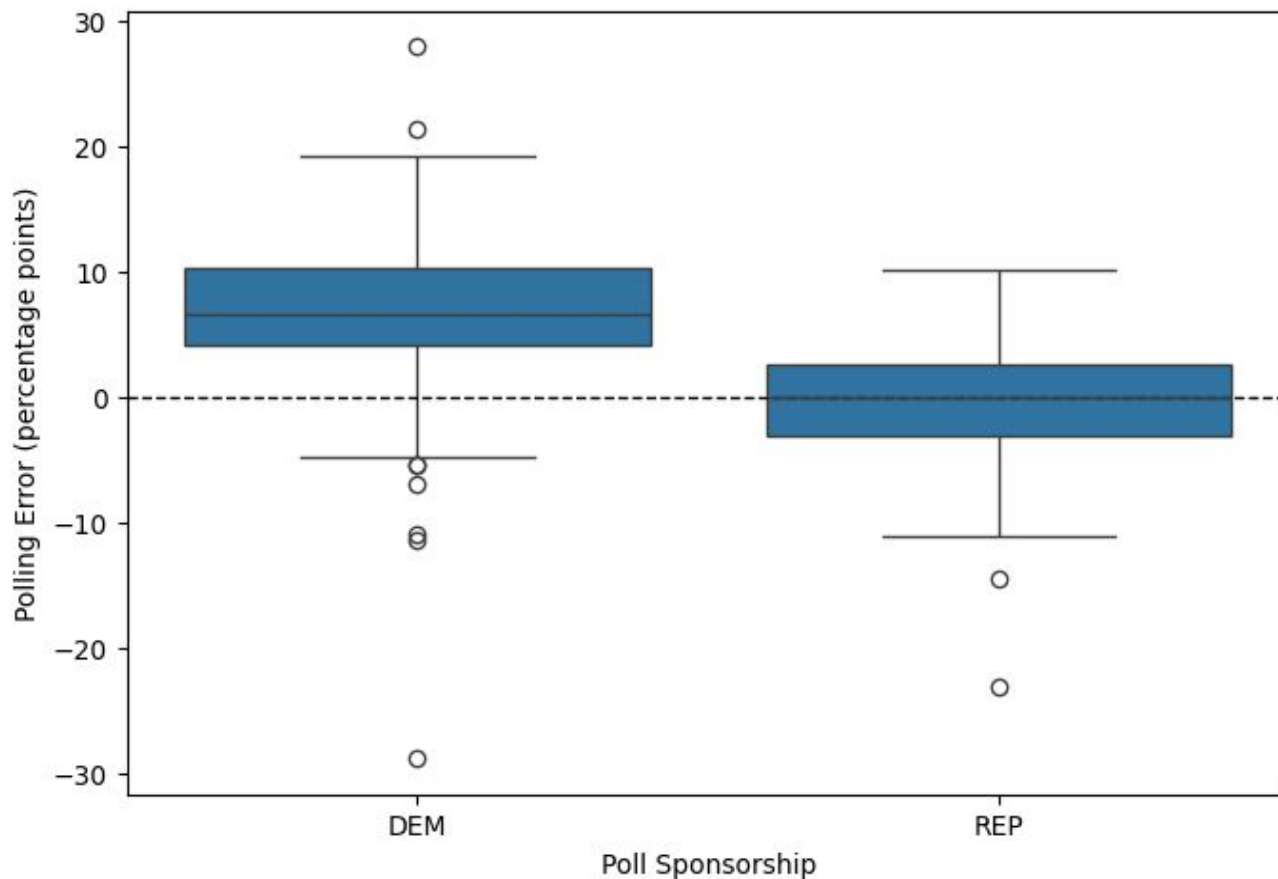
Partisan-sponsored polls are more evenly distributed across the campaign, while non-partisan polls cluster closer to Election Day.

2. Timing differences complicate comparisons of polling error

Because partisan and non-partisan polls are conducted at different times, raw error comparisons may conflate partisanship with campaign timing.

Kernel density estimates of poll timing by partisan sponsorship.

Partisanship Predicts Polling Error Even After Accounting for Timing



1. Polling error differs systematically by partisan sponsorship

The distribution of polling error varies substantially between Democratic- and Republican-sponsored polls.

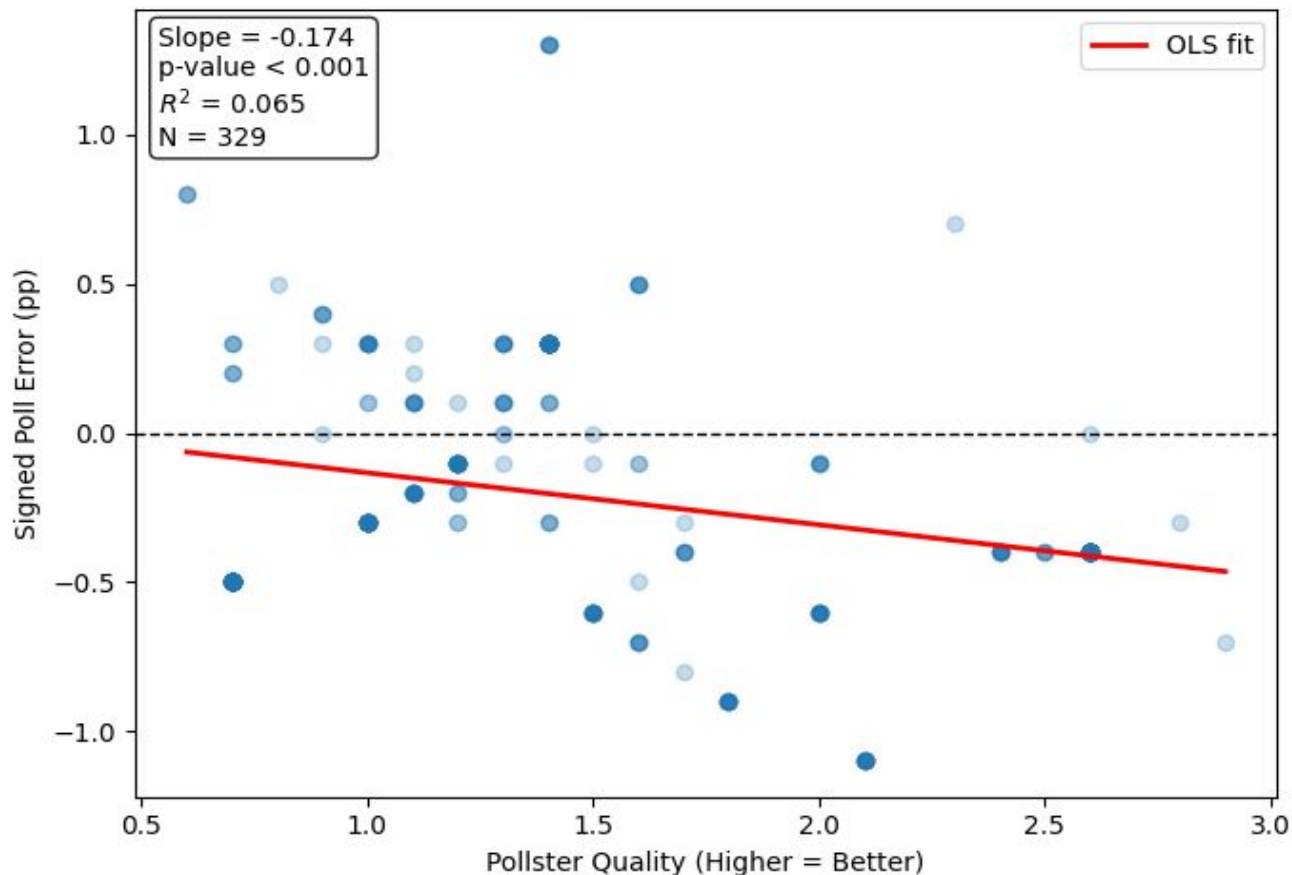
2. Errors differ in direction as well as magnitude

Democratic-sponsored polls tend to exhibit positive signed error, while Republican-sponsored polls tend to exhibit negative signed error and tended to be more accurate.

Poll sponsorship refers to the organization commissioning or releasing the poll; analysis includes Democratic- and Republican-sponsored polls only.

2. Pollster Quality

Polling Error Varies with Pollster Quality



1. Polling error varies systematically with pollster quality

Higher- and lower-rated pollsters do not exhibit identical error patterns. As p-value < 0.001, this is significant.

2. Higher-rated pollsters tend to underestimate outcomes

As pollster quality increases, signed polling error becomes more negative on average. This pattern is not intuitive, as higher-rated pollsters do not simply produce errors close to zero.

Pollster quality measured using FiveThirtyEight's numeric pollster ratings.

Pollster Quality Predicts Polling Error After Controls

Pollster Quality Coefficient

-0.17

$p < 0.001$

When conducting OLS regression of polling error

Number of Polls

329

After restricting to observations with complete data

A one-unit increase in pollster quality is associated with a **0.17 percentage-point shift toward more negative polling error**, conditional on timing, sponsorship, and sample size.

OLS Regression Equation

$$\text{error}_i = \beta_0 + \beta_1 \text{numeric_grade}_i + \beta_2 \text{time_to_election}_i + \beta_3 \text{samplesize}_i + \gamma_1 1(\text{DEM}_i) + \gamma_2 1(\text{REP}_i) + \varepsilon_i$$

If higher-rated pollsters also exhibit systematic error, this raises the possibility of election-wide dynamics rather than individual pollster failure.

3. Election-Wide Dynamics

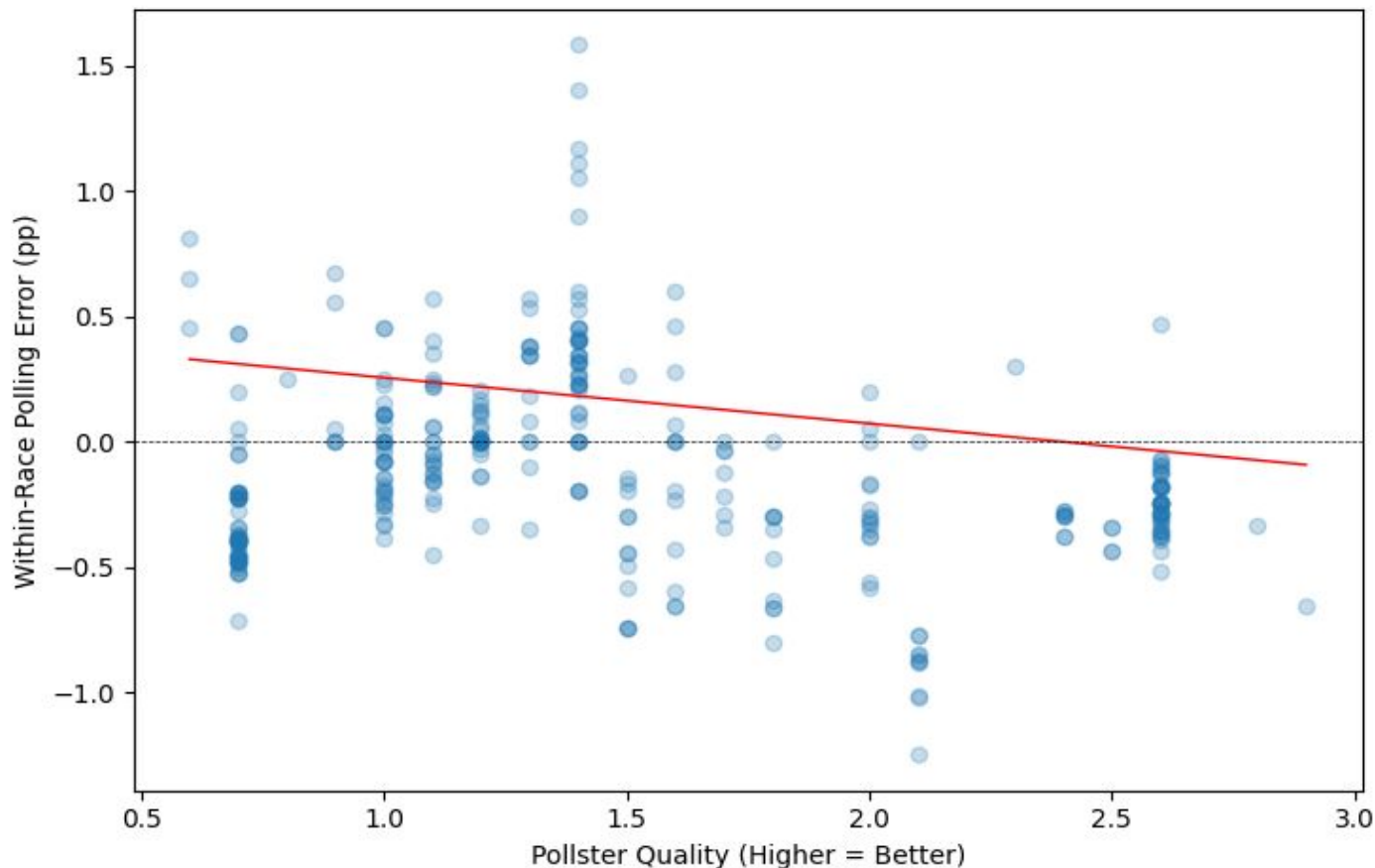
Higher rated pollsters do not meaningfully reduce the risk of large within-race polling errors

1. **Within-race polling error varies systematically with pollster quality**

Even after removing race-level average error, higher- and lower-rated pollsters do not exhibit identical error patterns ($p < 0.001$).

2. **Higher-rated pollsters tend to underestimate relative to peers**

As pollster quality increases, within-race polling error becomes more negative on average.



Pollster Quality Does Not Predict Typical or Tail Error

50th percentile coefficient

0.023

p = .346

When conducting quantile regression of
polling error

75th percentile coefficient

0.0171

p = .473

When conducting quantile regression of
polling error

While pollster quality predicts the direction of polling error, it **does not meaningfully reduce median or upper-tail error magnitude** once election-wide effects are removed.

Quantile Regression Equation

For quantile $\tau \in \{0.50, 0.75\}$:

$$Q_{\tau}(|\text{error}_i^{\text{demeaned}}| | X_i) = \beta_{0,\tau} + \beta_{1,\tau} \text{numeric_grade}_i + \beta_{2,\tau} \text{time_to_election}_i + \beta_{3,\tau} \text{samplesize}_i + \gamma_{1,\tau} 1(\text{DEM}_i) + \gamma_{2,\tau} 1(\text{REP}_i)$$

4. Future Directions

How can we get a better understanding of poll error?

Extending Analyses beyond 2020	Examining multiple election cycles would help distinguish election-specific dynamics from persistent pollster behavior.	1	Analyze whether the negative relationship between pollster quality and polling error observed in 2020 also appears in earlier cycles (e.g., 2012, 2016), or whether it reflects dynamics unique to 2020.
		2	Compare whether the quality–error relationship differs between state-level races (e.g., PA, WI, MI) and national presidential polls in 2020.
Disaggregate election-wide effects	Separating turnout shocks, late-breaking opinion shifts, and differential non response could identify the sources of shared race-level error.	1	Examine whether polling error was larger in races with unusually high turnout or shifts in absentee/mail voting, which were prominent in 2020.
		2	Test whether races with significant late events (e.g., COVID-19 surges, Supreme Court vacancy, economic news) exhibit stronger shared polling error across pollsters.
Evaluate pollster performance under different electoral conditions	Comparing presidential, Senate, and House races may reveal whether polling error behaves differently across contest types.	1	Compare pollster accuracy in close races versus safe races to see whether quality matters more under high electoral uncertainty.
		2	Evaluate whether the same pollsters systematically over- or underperformed across different states in 2020, indicating persistent pollster-specific tendencies.

→ Thank you!